B.Tech. Computer Science and Financial Technology

COURSE PLAN: THEORY COURSE

Department :	School of Computer Engineering			
Course Name & code :	Operating Systems CSF 3125		Core	
Semester & branch :	V Sem		Computer Science and Financial Technology	
Name of the faculty :	Renuka A			
No of contact hours/week:	L	T	P	C
	3	0	0	3

Course Outcomes (COs) to PO, PSO, BL Mapping

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	PSOs	BL *(Recommended)*
CO1	Summarize the operating system functions, services and structure	7	18	PO1, PO12	PSO1	L2, L3
CO2	Compare the various process scheduling and disk scheduling algorithms	8	22	PO2,PO4,PO12	PSO1	L3,L4
CO3	Apply the various methods for synchronization and deadlock prevention	7	20	PO3,PO4, PO12	PSO1, PSO3	L3, L4
CO4	Analyse the various memory management techniques	7	20	PO2,PO12	PSO1	L3,L4
CO5	Explain the concepts of file system and protection	7	20	PO1,PO12	PSO1, PSO3	L3
	Total	**36**	**100**			

CO	Engineering knowledge	Problem analysis	Design/development of solutions	Investigations of complex problems	Modern tool usage	Engineer and society	Environment and sustainability	Ethics	Individual and team work	Communication	Project management and finance	Life-long learning	Apply the knowledge in the domain of fintech	Build hands-on skills to use financial data	Develop fintech applications considering the ethical and legal
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3											1	3		
CO2		3		1								1	3		
CO3			3	1								1	3		2
CO4		3										1	3		
CO5	3											1	3		1
Average Articulation Level	3	3	3	1								1	3		1.5

ICT Tools used in delivery and assessment

Sl. No	Name of the ICT tool used	Details of how it is used
	Powerpoint slides	As a teaching aid
	LMS	For conducting assigenemnts
	MS Teams	For sharing resources

Typical tools including LMS, Smart Boards, MS Teams, etc

Mapping of Course Outcomes (COs)/Course Learning Outcomes (CLOs) #

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	Learning Outcomes (LOs)	BL *(Recommended)*
CLO1	Summarize the operating system functions, services and structure					
CLO2	Compare the various process scheduling and disk algorithms					
CLO3	Apply the various methods for synchronization and deadlock prevention					
CLO4	Analyse the various memory management techniques					
CLO5	Explain the concepts of file system and protection					
	Total					

 # Applicable to IET Accredited Courses (modules) Only

Delivery and Assessment Plan of LOs #

Learning Outcome (LO) mapped to the course		Delivery and assessment Plan
LO	**LO statement**	

 # Applicable to IET Accredited Programs Only

Assessment Plan (As communicated from o/o AD-A, in every odd semester)

IN – SEMESTER ASSESSMENTS									
Sl. No.	**Assessment Mode**		**Assessment Method**	****Time Duration**	****Marks**	****Weightage**	**Typology of Questions *(Recommended)***	****Schedule**	****Topics Covered**
1	**MISAC**	**1**	**Quiz**	Based on the complexity of the question		Marks based on the complexity of the question Total 5 marks	L2, L3	11 Aug 2025 – 19 Aug 2025	L1-L9
		2	**Quiz**	Based on the complexity of the question		Marks based on the complexity of the question Total 5 marks	L3	01 Sep 2025 – 08 Sep 2025	L10-L18
		3	**Mid-Term Test**			**Descriptive:** Marks based on the complexity of the question Total 30 Marks	L2, L3, L4	12 Sep 2025 – 18 Sep 2025	L1-L20
		4	**Quiz**	Based on the complexity of the question		Marks based on the complexity of the question Total 5 marks	L4	13 Oct 2025 – 21 Oct 2025	L18-L25

		5	**Quiz**	Based on the complexity of the question		Marks based on the complexity of the question Total 5 marks	L3	03 Nov 2025 – 08 Nov 2025	L25-l34
END – SEMESTER ASSESSMENT									
1	**Regular/Make–Up Exam**			180 Mins	50	Answer all 5 full questions of 10 marks each. Each question can have 3 parts of 2/3/4/5/6 marks.	L3, L4	17 Nov 2025- 29 Nov 2025	L1- L36

Lesson Plan

L No	Topics	CO Addressed
0	*To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.*	
1	What Operating Systems Do, Operating System Structure.	CO1
2	Operating System Operations, Process Management	CO1
3	Management, Storage Management. Operating System Service	CO1
4	User and Operating System Interface, System Call	CO1
5	Types of System Calls, System Programs	CO1
6	Operating System Structure, Virtual Machines, System Boot	CO1
7	Processes : Overview, Process Scheduling	CO2
8	Operations on Processes, Interprocess Communication	CO2
9	Threads: Overview, Multicore Programming, Multithreaded Models, Thread Libraries	CO2
10	CPU Scheduling: Basic Concepts, Scheduling Criteria	CO2
11	Scheduling Algorithms	CO2
12	Scheduling Algorithms-Examples	CO2
13	Thread Scheduling, Process Synchronization: Background	CO2
14	The Critical Section Problem, Peterson's Solution	CO3
15	Synchronization hardware, Mutex Locks, Semaphores	CO3
16	Classical Problems of Synchronization	CO3
17	Deadlocks : System Model, Deadlock Characterization	CO3
18	Methods for handling deadlock	CO3
19	Deadlock prevention	CO3
20	Deadlock Avoidance	CO3
21	Main Memory: Logical Versus Physical Address Space	CO4
22	Swapping, Contiguous Memory Allocation	CO4
23	Segmentation, Paging, Structure of Page Table	CO4
24	Virtual Memory :Background, Demand Paging	CO4
25	Copy-On-Write, Page Replacement	CO4
26	Allocation of Frame	CO4
27	Thrashing	CO4
28	Disk Scheduling:	CO2
29	Swap-Space Management	CO2
30	File Concept	CO5
31	Access Methods:	CO5
32	Directory and Disk structure ,File Sharing	CO5
33	File System Mounting	CO5
34	Protection:Goals of protection and Principle of protection	CO5
35	Domain of Protection,Access Matrix	CO5
36	Implementation of Access matrix	CO5

Faculty members teaching the course (if multiple sections exist):

Faculty	Section	Faculty	Section
Renuka A			

References:

Textbooks	• A. Silberschatz, P. B. Galvin and G. Gagne, Operating System Concepts, (9e), Wiley and Sons (Asia) Pte Ltd, 2016 • D.M. Dhamdhere, Operating Systems A Concept based Approach,(2e), Tata McGraw-Hill 2006 • Deitel & Deitel , Operating systems, (3e), Pearson Education, India. 2018. • Neil Matthew, Richard Stones, Beginning Linux Programing(4e), Wrox, 2007 • Andrew S Tanenbaum, Herbert Bos, Modern Operating systems. Pearson Education India, 2018
Self-Directed Learning	
Research Literature/ Case Studies	• •
NPTEL/Coursera/any MOOC-based material	• **noc25-cs141** •

Submitted by: Dr. Renuka A

(Signature of the faculty)

Date: 21/07/2025

Approved by:

(Signature of HoD)